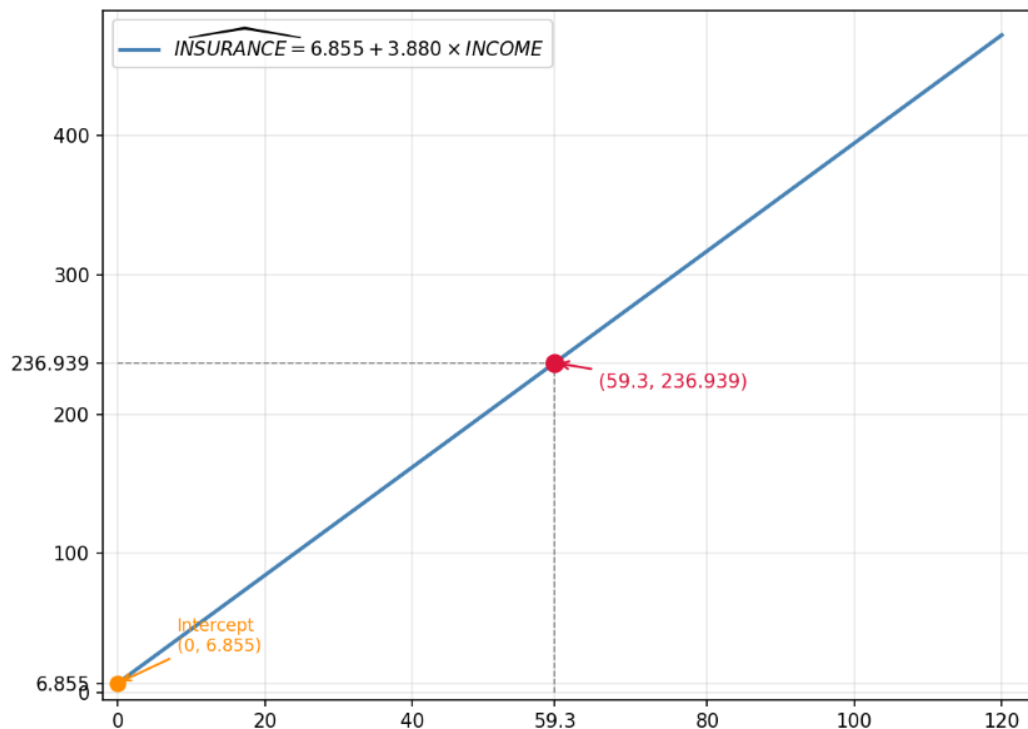


Problem 3.18

(a)



(b)

Point estimate: $b_2 = 3.880$ (thousands of dollars per thousand dollars of income).

95% Confidence Interval for β_2 ($df = N - 2 = 18$, $t_{0.025, 18} = 2.101$):

$$b_2 \pm t_{0.025, 18} \times se(b_2) = 3.880 \pm 2.101 \times 0.112 = 3.880 \pm 0.235$$

95% CI: (3.645, 4.115)

We are 95% confident that for every additional \$1,000 of household income, the average amount of life insurance held increases by between \$3,645 and \$4,115

(c)

For $x_0 = 100$ (\$100,000 income): $\hat{y} = 6.855 + 3.880 \times 100 = 394.855$

Variance of the fitted value:

$$\begin{aligned} \text{Var}(\hat{y}) &= \text{Var}(b_1) + x_0^2 \cdot \text{Var}(b_2) + 2x_0 \cdot \text{Cov}(b_1, b_2) \\ &= (7.383)^2 + (100)^2(0.112)^2 + 2(100)(-0.746) \\ &= 54.509 + 125.440 - 149.200 = 30.749 \end{aligned}$$

$$se(\hat{y}) = \sqrt{30.749} \approx 5.545$$

99% CI ($t_{0.005, 18} = 2.878$):

$$\hat{y} \pm t_{0.005, 18} \times se(\hat{y}) = 394.855 \pm 2.878 \times 5.545 = 394.855 \pm 15.957$$

99% CI: (378.898, 410.812)

(d)

Hypotheses:

$$H_0: \beta_2 = 5$$

$$H_1: \beta_2 \neq 5$$

Test statistic:

$$t = (b_2 - 5) / \text{se}(b_2) = (3.880 - 5) / 0.112 = -1.120 / 0.112 = -10.000$$

Rejection region: $|t| > t_{0.025, 18} = 2.101$

Since $|-10.000| = 10.000 > 2.101$, reject H_0 at the 5% level of significance.

Conclusion: The data do not support the claim that each \$1,000 increase in income raises average insurance by \$5,000.

(e)

Hypotheses:

$$H_0: \beta_2 = 1$$

$$H_1: \beta_2 > 1$$

Test statistic:

$$t = (b_2 - 1) / \text{se}(b_2) = (3.880 - 1) / 0.112 = 2.880 / 0.112 = 25.714$$

Rejection region: $t > t_{0.01, 18} = 2.552$

Since $25.714 > 2.552$, reject H_0 at the 1% level of significance.

Conclusion: There is sufficient evidence that the slope $\beta_2 > 1$, insurance increases by more than the same amount as income.